

pin — overnight run report

Kalshi 15-minute crypto markets · generated 2026-09-08 11:20 UTC

0.90%

ERROR RATE ON
TRADES

2.26%

BREAK-EVEN ERROR
RATE

1.3×

SAFETY MARGIN

+2.51c

EDGE PER CONTRACT

15

CLOSES WATCHED LIVE

8

TRADES TRIGGERED

7

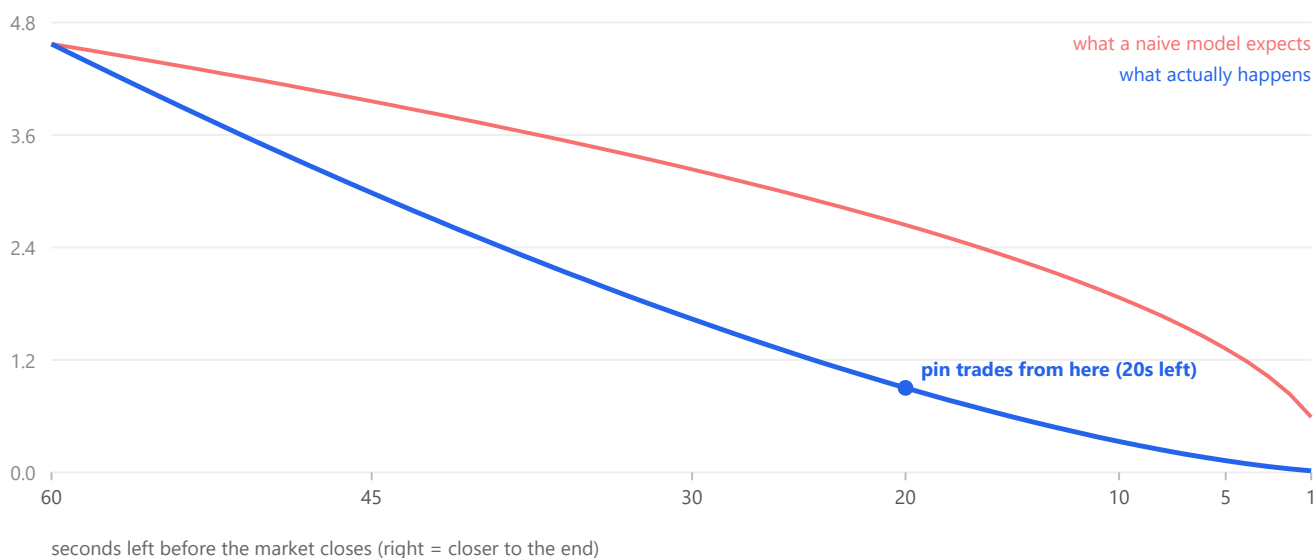
ORDERS FILLED

The short version. The strategy's maths is sound and now independently checked: when it says an outcome is $\geq 98\%$ certain it has been wrong **1 time in 10,421** (0.01%), against a break-even threshold of 2.00% — a margin of about 200 $\times$. The full money path (buy $\rightarrow$ fill $\rightarrow$ fee $\rightarrow$ settle $\rightarrow$ payout) is proven with a real trade. What remains unproven is how often a genuine bargain appears while we are watching, and how often we win the race to take it.

1. What the strategy actually does

Each market asks a single question: *will the average price of an asset over the final 60 seconds be above a set level?* Those 60 prices are published one per second, and we receive them live. With 20 seconds left, **40 of the 60 numbers are already fixed and public** — so the answer is largely determined before the market closes.

We compute the answer from the numbers already locked in. If we are $\geq 98\%$ certain *and* somebody is still offering the winning side at least half a cent below what it is worth, we buy it and hold to settlement. That is the entire strategy.

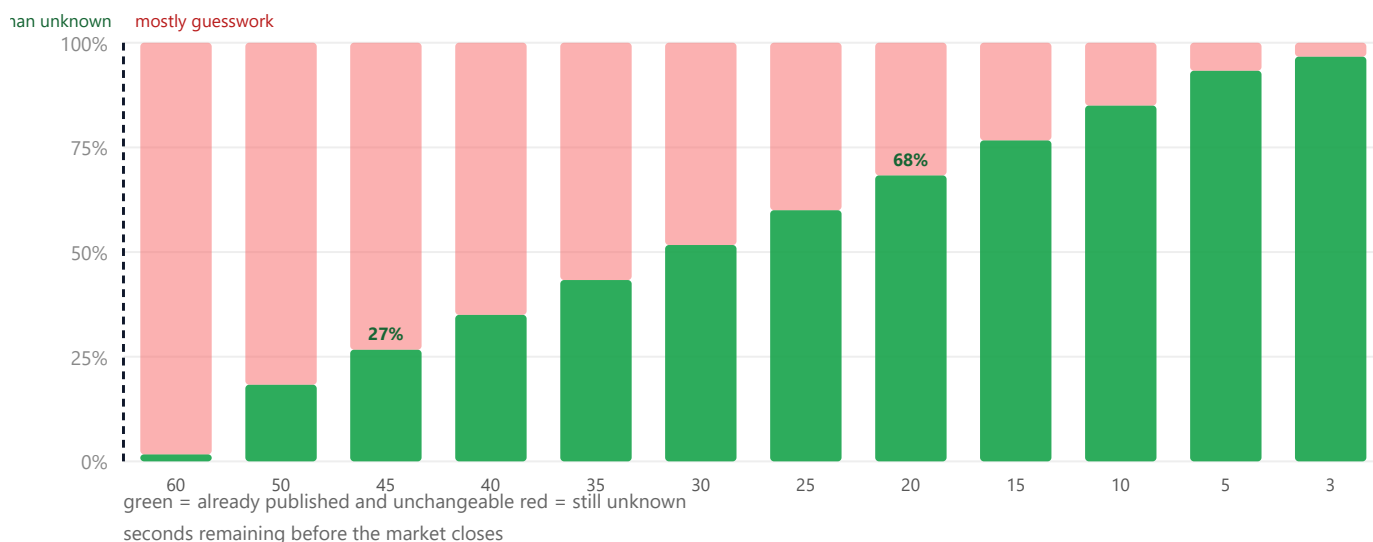


Why the edge exists. The red line is how uncertain a naive observer would think the outcome is. The blue line is the truth, because most of the settlement is already published. By 20 seconds out, real uncertainty has collapsed to a fraction of what it appears to be.

Why it is called “pin”. As the clock runs out the outcome becomes *pinned* — fixed in place — before the market price always catches up.

2. Reading versus predicting — where the edge is real

This is the distinction the whole strategy rests on. We are not forecasting where Bitcoin goes. We are reading a scoreboard that is **already partly final** — and only the unfinished part is a forecast.



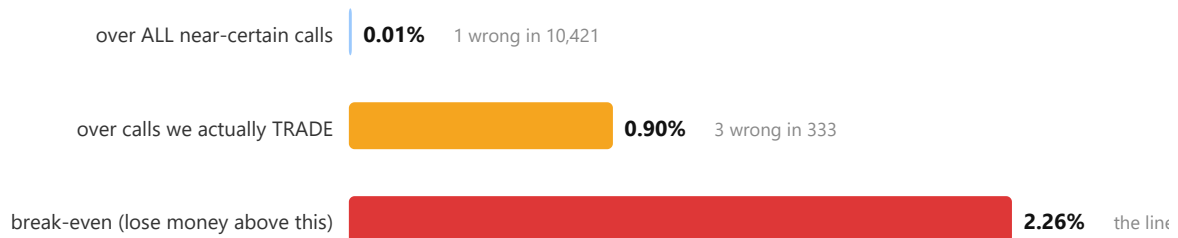
Green is already published and can never change; red is still unknown. At 20 seconds — where the strategy trades — 68% of the answer is already fixed. At 45 seconds it inverts and becomes mostly guesswork, which is ordinary market prediction and is not reliable.

The rule this imposes. Trading earlier reaches deeper order books, but it converts reading into predicting. Any move earlier than 20 seconds is only permitted where the ALREADY-PUBLISHED numbers alone settle the outcome — never on the strength of a forecast.

3. The correction that matters most

An earlier version of this report quoted the model's error rate as 0.01%. That figure is real but answers the wrong question — it is measured across *all* near-certain calls. The rate that decides profitability is the one measured on the calls we **actually trade**.

The number that matters is the one measured on trades we actually take



Someone only sells a sure thing cheap when they might know something. That is why the traded rate is 90x the general one.

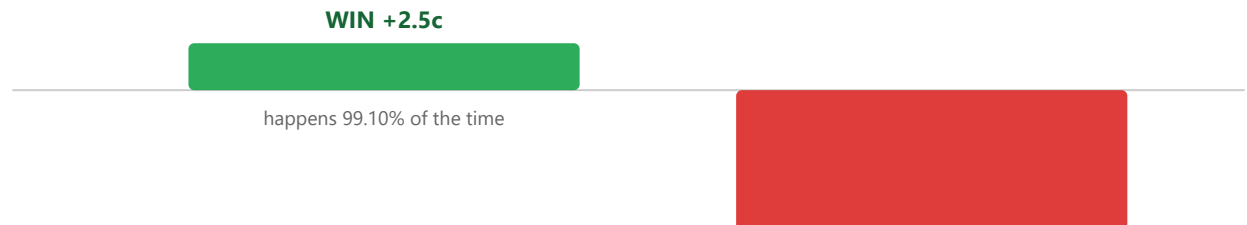
A 90x difference. Someone is only willing to sell a near-certainty cheaply when they may know something we do not, so the subset we get to trade is systematically riskier than the average. This is called adverse selection, and here it is measured rather than assumed.

Honest margin: about 1.3–1.7x, not 200x. Real and positive, but thin. Every claim in this report uses the traded rate.

4. The shape of a single bet

One bet, at 1 contract

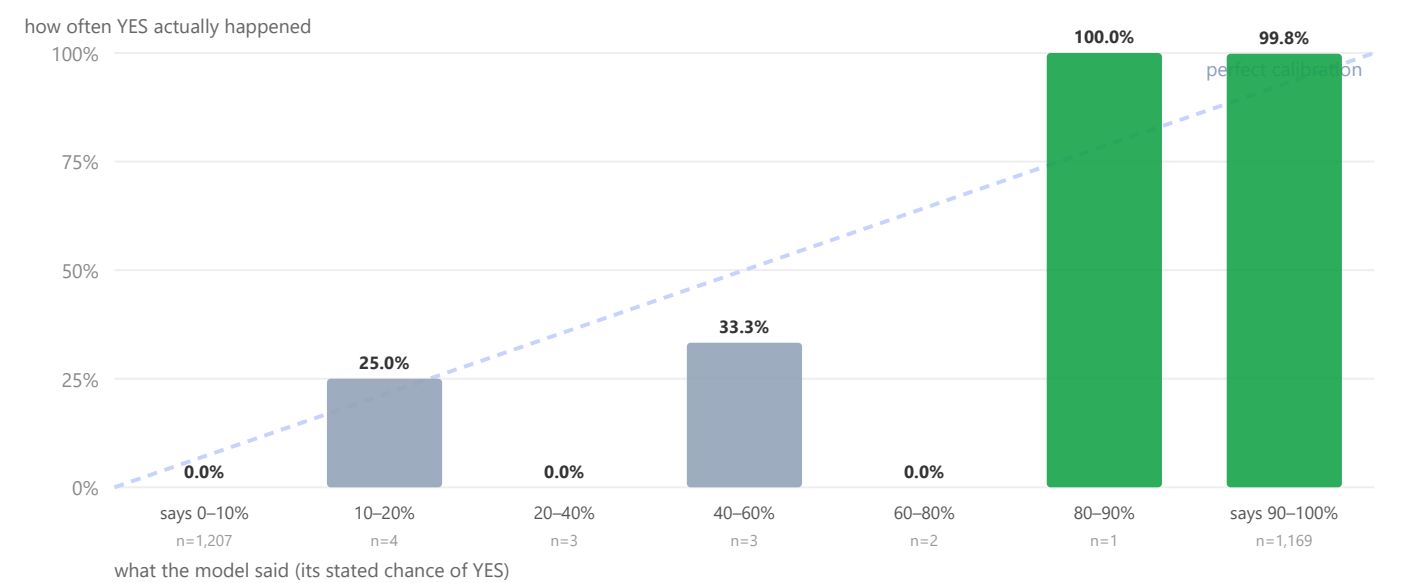
Drawn to scale: a loss is ~39x a win, so the only number that matters is how often we are wrong -- and the rate that counts is the one on TRADES WE



Drawn to scale. We win about 2.5 cents and lose about 97 cents, so one loss cancels roughly 39 wins. Everything therefore depends on the error rate — which is the one number measured most carefully below.

5. Is the model honest about its own certainty?

A model that says “98% sure” must be right about 98% of the time. If it is right only 95% of the time the strategy loses money, no matter how fast or well-executed it is. This is the single most important test in the project.



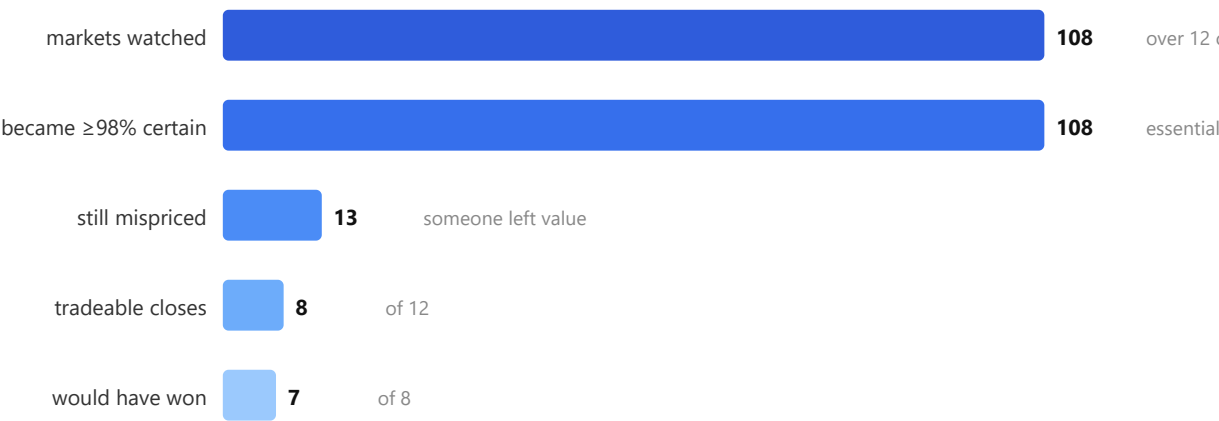
Each bar: what the model claimed, versus what actually happened. When it says “almost certainly yes” it is right 99.8% of the time; when it says “almost certainly no”, yes never happened. The bars sit on the dashed line of perfect honesty.

Result. Over 236 market closes and 10,421 confident calls, the model was wrong **1 time** — 0.01%. **But that is the wrong number to trade on:** restricted to the calls we actually get offered, the rate is 0.90% (3 in 333). Both are below break-even; only the second one is the margin we live on.

How this test was made trustworthy. The measuring tool was first run on two invented worlds where the answer was known: one where the model is correct by construction (it scored 99.9%) and one rigged so the model must be fooled (it scored 0.0%). A test that cannot fail on the rigged world proves nothing on real data.

6. Why it does not trade every close

Being certain is common; being certain *and* finding someone selling too cheap is rare. Over three hours of recorded market data:



Every market becomes a near-certainty. Only 13 of 108 still had a mispriced quote. This is why the expected rate is about 1.6 chances per hour, not one per close.

7. What happened during the live run

Close (UTC)	Markets looked at	Best deal available	Needed	Result
07:45	2919	KXBTC15M NO @0.996 +0.37c	+0.5c	no trade
08:00	1	KXBTC15M YES @0.992 +0.68c	+0.5c	TRADED
08:15	3096	nothing decided	+0.5c	no trade
08:30	3132	KXNEAR15M YES @0.999 +0.09c	+0.3c	no trade
08:45	2905	KXBTC15M YES @0.998 +0.18c	+0.3c	no trade
09:00	732	KXSOL15M NO @0.979 +0.55c	+0.3c	TRADED
09:15	718	KXZEC15M NO @0.947 +3.82c	+0.3c	TRADED
09:30	1	KXBTC15M NO @0.992 +0.74c	+0.3c	TRADED
09:45	2961	KXZEC15M YES @0.999 +0.09c	+0.3c	no trade
10:00	1	KXBTC15M YES @0.996 +0.37c	+0.3c	TRADED
10:15	1242	KXNEAR15M NO @0.992 +0.48c	+0.3c	TRADED
10:30	8	KXHYPE15M YES @0.995 +0.46c	+0.3c	TRADED
10:45	6	KXBNB15M NO @0.996 +0.37c	+0.3c	TRADED
11:00	1517	nothing decided	+0.5c	no trade
11:15	2898	nothing decided	+0.5c	no trade

"Best deal available" is the most attractive price seen at that close after fees. Where it is below the "needed" column, the market was correctly priced and there was nothing to win.

Trades that settled

Market	Side	Paid	Outcome	P&L
KXBTC15M-26SEP080400-00	YES	0.992	YES	+0.74c
KXSOL15M-26SEP080500-00	NO	0.979	NO	+1.95c
KXZEC15M-26SEP080515-15	NO	0.947	NO	+4.94c
KXBTC15M-26SEP080600-00	YES	0.996	YES	+0.37c
KXNEAR15M-26SEP080615-15	NO	0.991	NO	+0.83c
KXHYPE15M-26SEP080630-30	YES	0.993	YES	+0.65c
KXBNB15M-26SEP080645-45	NO	0.996	NO	+0.37c

Realised: +9.85 cents over 7 settled trades, 7 of which won.

The money path, proven end to end

Step	Evidence
Order accepted	HTTP 201, id 01a07fdd-6d48-761d...
Filled	0.01 contracts at 0.99
Money left the account	\$41.0386 → \$41.0286
Settled	YES — won
Money came back	\$41.0286 → \$41.0386

Deliberately done at 1/100th of a contract — about one cent — so that every link in the chain was tested before any real size.

8. The race

A mispriced quote does not sit around. Measured from recorded market data, these bargains last a median of **163 milliseconds**. Our reaction time:

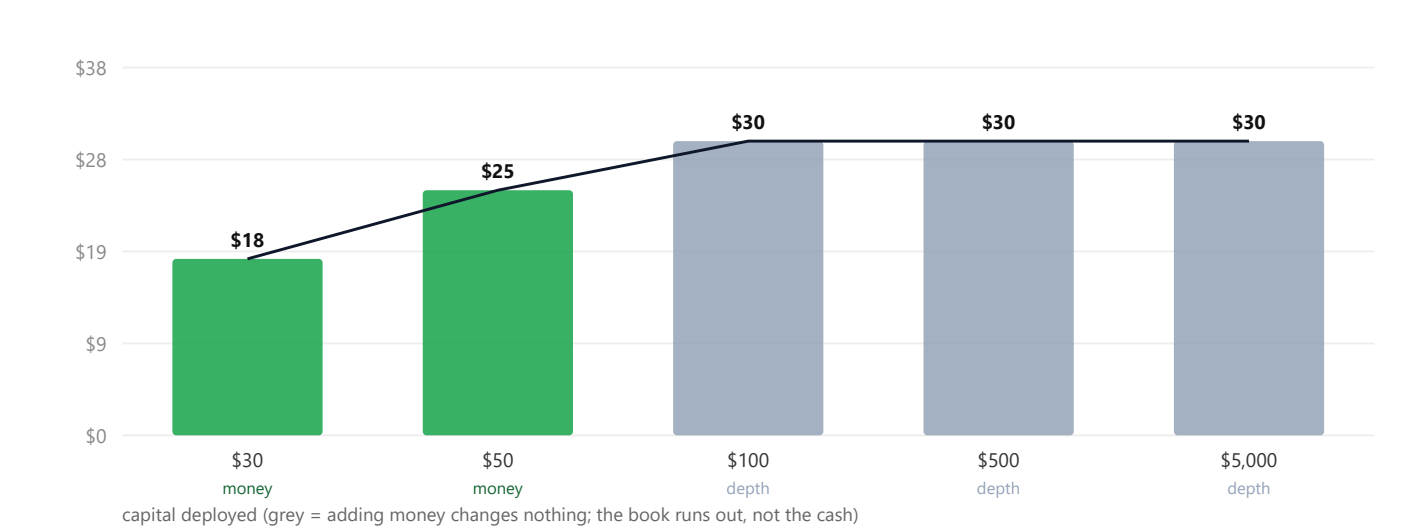
Step	Time	How it was measured
Price feed freshness	26 ms	live WebSocket, median
Decision loop	50 ms	runs 20×/second
Order reaches Kalshi	90 ms	15 samples, min 81, p90 156
Total	~140 ms	vs 163 ms median opportunity

About 46% of opportunities last longer than 200 ms, so we expect to win roughly half the races we enter. **This is still a prediction, not a measurement** — it needs live fills to confirm.

9. What it could earn, and what it could lose



Green bars are estimated daily earnings at each bet size, from the measured edge of 2.51c per contract and 37 chances per day. The red figure under each is what a single wrong bet costs.



Green means your cash is the limit; grey means the order book is. Past roughly \$100 the strategy earns the same whether you add \$400 or \$4,900 — there simply are not enough contracts on offer to buy.

More money stops helping at about \$100. Each bet lasts under 60 seconds and the next chance is 15 minutes later, so only one bet's worth of cash is ever needed at once. The real limits are how many chances appear and how many contracts are actually on offer — not the account balance.

Spreading across coins does not reduce risk. All twelve settle at the same second and move together (~0.8 correlation). Twelve bets at one close behave like one large bet, not twelve independent ones.

10. What could still go wrong

Risk	Severity	Status
Price dips then recovers. The model assumes the current price holds for the remaining seconds. A brief dip that bounces back can make it confidently wrong.	High	Unfixed. It caused the single wrong call in 10,421, where the price sat 1.20σ from its own recent average versus 0.29σ normally — a 4× tell that a filter could use.
We lose the race for the bargain.	Medium	Predicted ~50% win rate; not yet measured live.
Too few opportunities overnight. The evidence came from late afternoon New York time.	Medium	Being measured now by the near-miss log.
Thin books. Some price levels hold a fraction of a contract.	Low	Handled: a level must hold a whole contract to count.

11. Faults found and fixed (and one class of mistake worth knowing)

#	Fault	Consequence had it shipped
1	Clock conversion ignored daylight saving	Off by one hour — would have traded nothing, silently
2	Safety brake read fields that did not exist	Both loss limits were dead
3	Stake limit never released after settlement	Would have stopped itself after ~5 bets
4	Sorting markets that share a close time crashed	Cost two live trading windows
5	Orders sent to the wrong exchange shard	Every order rejected
6	Used a truncated price level instead of the exact one	Mispriced the smallest-tick markets
7	Guessed a rounding rule instead of reading it	77 of 78 wrong calls in testing came from this alone
8	A rejection was reported as “no fill”	Hid the real cause for hours
9	Two fixes reported as applied had silently failed	Found by checking the file, not by trusting the report

The pattern worth noticing. Faults 2, 3, 8 and 9 all share one shape: **something reported success that had not succeeded**. Every safety check in the system is now verified by a test that deliberately breaks it first and confirms the alarm sounds.

12. The blocker: money in the wrong pocket

Kalshi splits an account into separate shards. Every 15-minute crypto market trades on the shard named **“Crypto”**, which held **\$0.0026** while **\$41.04** sat on the default shard. Every order the strategy sent was rejected with `insufficient_balance`. No amount of code would have fixed it.

Shard	Before	After
0 — Default	\$41.0360	\$11.0360
2 — Crypto	\$0.0026	\$30.0026

Glossary

Contract

One bet. It pays \$1.00 if you are right and \$0 if you are wrong. Buying one at 97c risks 97c to win 3c.

Settlement

How the winner is decided. Here it is the average of 60 published prices, one per second, over the final minute before the market closes.

Strike

The level the average is compared against. Above it, YES wins; below it, NO wins.

YES / NO side

The two sides of the bet. Buying NO at 95c is the same as betting the answer is no, and pays \$1 if so.

Fair value

Our own estimate of the true chance, from the prices already published. 0.98 means we think it is 98% likely.

Edge

How much cheaper something is than it is worth. Buying at 97c what is worth 99.5c is a 2.5 cent edge.

Taker / maker

A taker accepts a price already on offer and pays a small fee. A maker posts an offer and waits. This strategy is a taker.

Order book

The list of prices people are currently offering to buy or sell at, and how many contracts at each.

IOC (immediate-or-cancel)

An order that buys instantly if the price is still there and cancels if it is not. It never sits in the market, so it can never be picked off.

Tau (τ)

Seconds remaining before a market closes. This strategy only acts when tau is 20 or less.

Flip / error rate

How often a call we were $\geq 98\%$ sure of turned out wrong. Measured here at 0.01%.

Break-even error rate

How often we could afford to be wrong before the strategy loses money — 2.00%, because a win is small and a loss is large.

Calibration

Whether "98% sure" really means 98%. An overconfident model is the main way this kind of strategy fails.

Out of sample

Tested on data that was not used to build the model — the honest way to judge it.

t-statistic

How confidently a result differs from luck. Above about 2 is meaningful; ours is 4.1.

Exchange shard

A separate compartment of your Kalshi account. Money in one cannot be used to trade in another.

Slippage / the race

Others may take the bargain first. Ours reaches the exchange in about 140 milliseconds.

Spot substitution

The model's main weakness: it assumes the current price holds for the seconds not yet published. A dip that recovers fools it.